MC Question	Lec0101 Answer	Lec0201 Answer
1	c	c
2	b	c
3	d	d
4	e	b
5	a	b
6	a	a
7	e	e
8	d	b
9	e	e
10	d	b

Short Answers for Lec0101

Q11:

Answers may vary. Sample answer:

```
sample(x=StudentID, size=30, replace=TRUE)
```

For full points:

- must use StudentID as the selection argument (as other two variables are not unique) or if used something else, must explain how they linked whatever vector is used to the dataframe or must explain why that other variable is unique to each student.
- must include 30
- must be sampled with replacement

Q12:

Answers may vary. Sample answer:

Impact: patients not in the study may see less benefit if the research is not reproducible.

Right: Privacy of patients in the study

Comment: In order to protect the privacy rights of patients in the study we should not publish any summary statistics which could lead to the identification of any patient.

Q13:

$$E(\bar{Y}^2) = Var(\bar{Y}) + E(\bar{Y})^2$$
$$E(\bar{Y}^2) = \frac{\sigma^2(N-n)}{n(N-1)} + \mu^2$$
$$E(n\bar{Y}^2) = nE(\bar{Y}^2)$$
$$E(n\bar{Y}^2) = \frac{\sigma^2(N-n)}{(N-1)} + n\mu^2$$

Short Answers for Lec0201

Q11.

Answers may vary. Sample answer:

```
age <- rpois(n=238, lambda=30)
```

For full points:

- results from simulation must be discrete.
- results from simulation must be non-negative.
- There must be 238 simulated values, which come from a population with mean of 30 and population variance of 30.

Q12:

Answers may vary. Sample answer:

Impact: students not in the study may see less benefit if not all data is collected/analyzed.

Right: Students in the study have the right to not answer questions or “leave” the study.

Comment: To protect the privacy rights of students in the study we should make questions optional.

Q13:

$$E(\bar{Y}^2) = Var(\bar{Y}) + E(\bar{Y})^2$$
$$E(\bar{Y}^2) = \frac{\sigma^2(N-n)}{n(N-1)} + \mu^2$$
$$E((N-1)\bar{Y}^2) = (N-1)E(\bar{Y}^2)$$
$$E((N-1)\bar{Y}^2) = \frac{\sigma^2(N-n)}{n} + (N-1)\mu^2$$